

Md. Almabud **JUWAL**

Senior Software Engineer | B.Sc. Computer Science and Engineering

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 Dhaka, Bangladesh

 EDUCATION

2020	B.Sc. in Computer Science and Engineering
2016	> Daffodil International University, Dhaka, Bangladesh > CGPA : 3.77 / 4.00 > Relevant coursework : Artificial Intelligence, Algorithms, Data Structures, Software Engineering, Compiler Design, Discrete Mathematics, Computer Networks > Medium of instruction : English

 ACADEMIC & RESEARCH INTERESTS

INTERESTS : Artificial Intelligence, Machine Learning, Data Science,
Software Engineering, Distributed Systems

 SKILLS

Programming	Python, SQL, JavaScript/TypeScript, Go, C/C++
Backend & Frameworks	FastAPI, Django, Django REST Framework, Celery
Data & Databases	PostgreSQL, Pandas, Redis, MongoDB, Neo4j, Elasticsearch
Infrastructure & Engineering	Docker, RabbitMQ, Git, CI/CD, Jenkins, Ansible
Frontend	Angular

 EXPERIENCE

Present Jun 2024	Senior Software Engineer Cogwheel Analytics, Colorado, United States > Architected and led the development of the next-generation Cogwheel Analytics platform as a modular monolith following Clean Architecture principles, with clear separation of concerns and reusable handling for authentication, pagination, and request validation, reducing repetitive implementation work and accelerating feature delivery > Lead the development and maintenance of the Data Cleaning service, implementing data-validation and transformation workflows to improve data integrity and consistency > Design and implement backend services and APIs using Python, FastAPI, and PostgreSQL, with emphasis on maintainability, reliability, and performance > Build and maintain asynchronous data-processing and web-scraping workflows using Pandas, Celery, and Scrapy Python FastAPI PostgreSQL Pandas Celery Scrapy Angular
May 2024 Jul 2023	Senior Software Engineer Brainstation 23 Ltd, Dhaka, Bangladesh > Led the turnaround of a delayed project by mentoring 3 junior engineers, restructuring task allocation, and redesigning the system architecture, enabling successful delivery of the second version within 6 months > Designed software architectures for Task Station and Sayvery, with focus on scalability, maintainability, and clear separation of responsibilities > Managed the technical backlog by translating product and stakeholder requirements into prioritized engineering tasks aligned with project goals and delivery timelines > Designed and implemented web-scraping and invoice-parsing solutions to automate data collection and extraction Python Django Django REST Framework PostgreSQL Celery Scrapy

Jun 2023 Oct 2021	Software Engineer Brainstation 23 Ltd, Dhaka, Bangladesh ➤ Improved system performance by approximately 10% through database-query optimization ➤ Improved API response performance by approximately 5% by introducing application-level caching ➤ Upgraded Python and Django versions and refactored the existing codebase to improve maintainability and compatibility ➤ Contributed to real-time synchronization between backend and frontend systems using WebSockets and Go, while mentoring an Associate Software Engineer Python Django Django REST Framework PostgreSQL Redis Celery Go
Sep 2021 Jul 2020	Junior Software Engineer Tirzok Private Limited, Dhaka, Bangladesh ➤ Reduced processing time for 3.3 million records from approximately 29–35 minutes to 2–4 minutes using PostgreSQL COPY and database triggers ➤ Reduced database query response time by approximately 50% through query optimization ➤ Contributed to backend and database architecture, including the implementation of a custom authentication backend and graph-based data solutions using Neo4j ➤ Automated deployment workflows using Jenkins and Ansible, reducing manual deployment effort and improving release consistency Python Django PostgreSQL Redis Neo4j Jenkins Ansible Angular
Apr 2020 Dec 2019	Web Developer Wizdoor Inc, Dhaka, Bangladesh ➤ Contributed to backend and database architecture while coordinating implementation with frontend and design teams ➤ Developed a real-time chat application using Django Channels and WebSockets Python Django Django REST Framework PostgreSQL Django Channels Angular

PROJECTS

COGWHEEL ANALYTICS

Jun 2024 – Present

<https://cogwheelanalytics.com>

Enterprise analytics platform for automated data ingestion, validation, processing, and reporting. The next-generation platform is designed as a modular monolith following Clean Architecture principles, with reusable application components and asynchronous processing to support maintainable and scalable analytics workflows.

Python FastAPI PostgreSQL Pandas Celery RabbitMQ Scrappy Angular

REHIVE

Oct 2021 – Feb 2023

<https://rehive.com>

Contributed to a South Africa-based fintech platform supporting multiple currencies, including cryptocurrency. Implemented the billing service and contributed to a plugin-based architecture that enables additional services and commerce-oriented functionality to be integrated with the platform.

Python Django Django REST Framework PostgreSQL Celery RabbitMQ Redis Docker Angular

LEGISLIB

May 2020 – Jul 2020

Legal-document processing and search platform for case-law materials from Bangladesh, India, the UK, and the USA. Processed and reformatted PDF documents and indexed legal content to provide searchable access for Bangladeshi legal professionals and students.

Python Django Django REST Framework MongoDB Elasticsearch PDF Processing

ACADEMIC & PROGRAMMING ACHIEVEMENTS

2018	Participant, ACM ICPC Dhaka Preliminary Contest
2018	4th place , DIU Intra Team Programming Contest
2017	2nd place , DIU Intra Team Programming Contest
2017	9th place , DIU Takeoff Individual Programming Contest

CERTIFICATIONS & COURSES

Jun 2025	Advanced Certificate for Management Professionals (ACMP 4.0), University of Dhaka
Apr 2023	AWS Certified Cloud Practitioner
2021	Object-Oriented Design, Coursera
2021	The Ultimate Docker Course, Mosh Hamedani

TEACHING AND MENTORING

Dec 2018	ACM Co-Chair Computer & Programming Club, Daffodil International University > Coordinated programming contests and other ACM-related club activities > Mentored students in competitive programming and encouraged participation in programming contests
Jan 2018	
Jan 2019	Student Prefect Daffodil International University > Assisted students with Competitive Programming and Algorithms through practical lab sessions and academic support > Assisted the course instructor with laboratory examinations and student performance evaluation > Mentored students requiring additional academic guidance
May 2017	

LANGUAGES

- > **Bengali** — Native
- > **English** — B2 (CEFR), IELTS Academic 6.5
- > **Hindi** — Conversational